FLY ASH COMPREHENSIVE DOSSIER

Prepared: December 10, 2025 | **Version:** 2.0 | **Focus:** US Supply/Demand Net Balance Analysis

EXECUTIVE SUMMARY: The US fly ash market faces a structural supply deficit as coal plant closures accelerate (-48% capacity since 2011 peak) while infrastructure demand remains elevated through IIJA spending ($1.2T through 2028). Beneficial use rates reached record 72% in 2024 (10th consecutive year >50%), but total CCP production declined from 80M tons (2022) to 67M tons (2023). Import arbitrage remains attractive: Asian FOB $16-27/MT vs US $118-123/MT. Supply gap projected to widen 2025-2030 as 68.8 GW additional coal capacity retires.

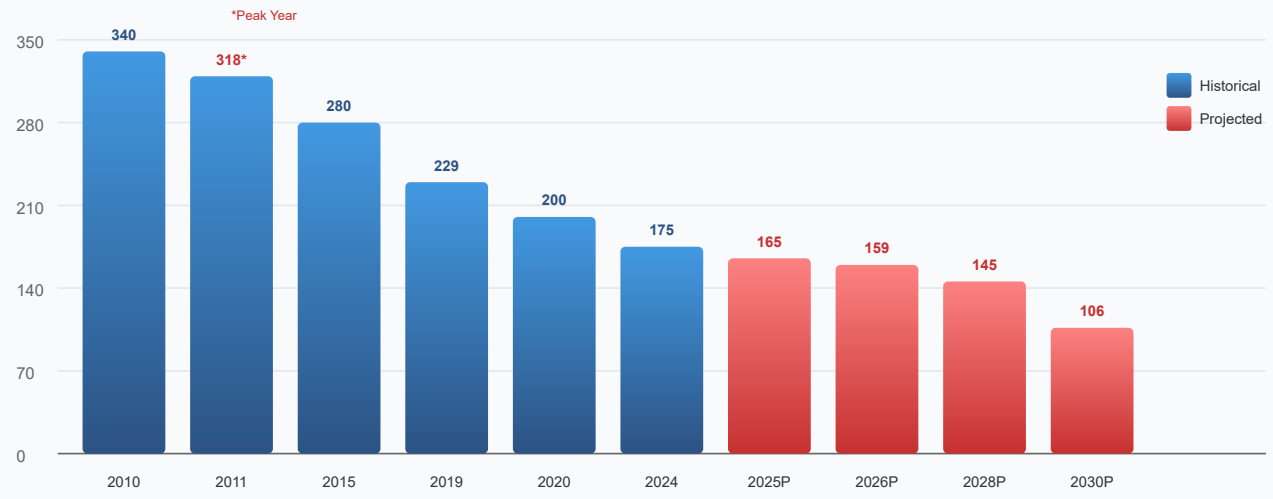
67M
Total CCP Produced (2023)
-16% from 2022

72%
Beneficial Use Rate (2024)
+10% from 2022

165
Coal Capacity GW (End 2025)
-48% from 2011 peak

$123
US Price per MT (2025)
+$5 from 2024

1. US COAL CAPACITY DECLINE - THE SUPPLY CRISIS



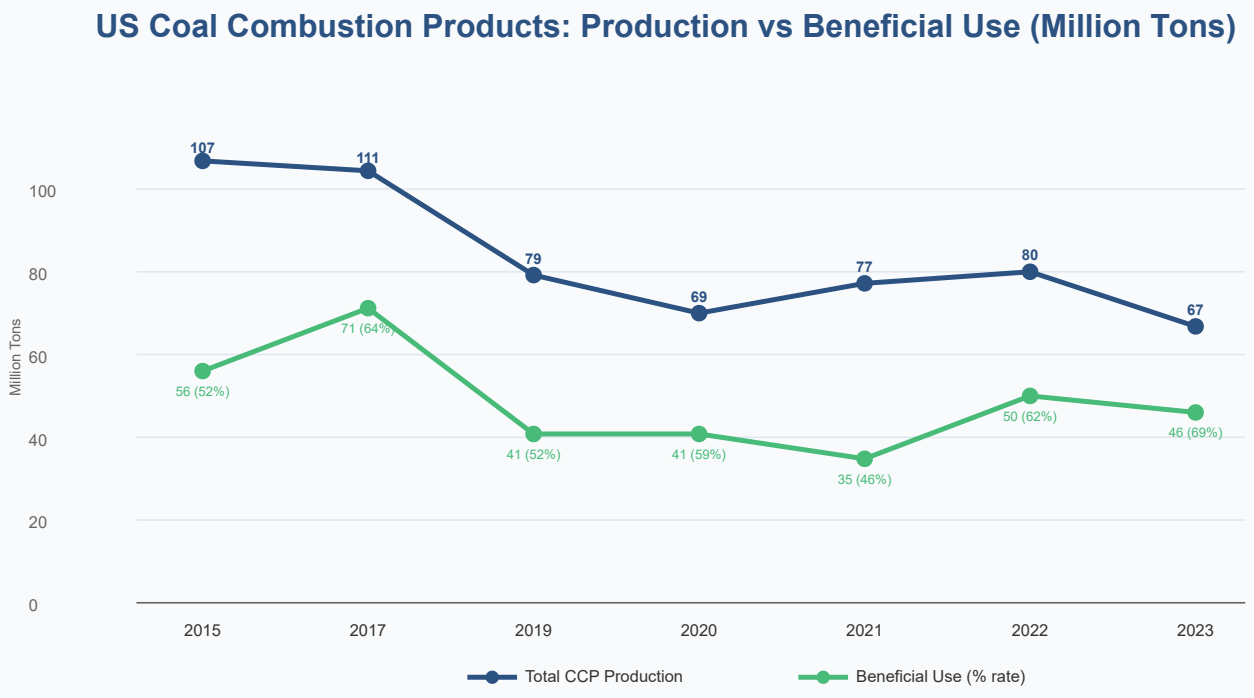
Sources: EIA, IEEFA. 2025-2030 projections based on announced retirements. 68.8 GW scheduled to retire 2025-2030.

1.1 Coal Plant Retirement Acceleration (2025)

Metric	2023	2024	2025 (Projected)	Change
Capacity Retired (MW)	10,577	3,629	14,532	+300% YoY
Coal Consumed by Retiring Units (M short tons)	28.5	9.0	39.6	+340% YoY
Primary Regions Affected	MISO (39%), PJM (27%), SPP (15%)			

2025 RETIREMENT SURGE: Record 14.5 GW retiring in 2025 - highest since coal decline began. Units consumed 39.6M short tons of coal in 2023. This removes significant fly ash production capacity from Midwest and Mid-Atlantic regions (MISO + PJM = 66% of retirements).

2. FLY ASH PRODUCTION vs BENEFICIAL USE



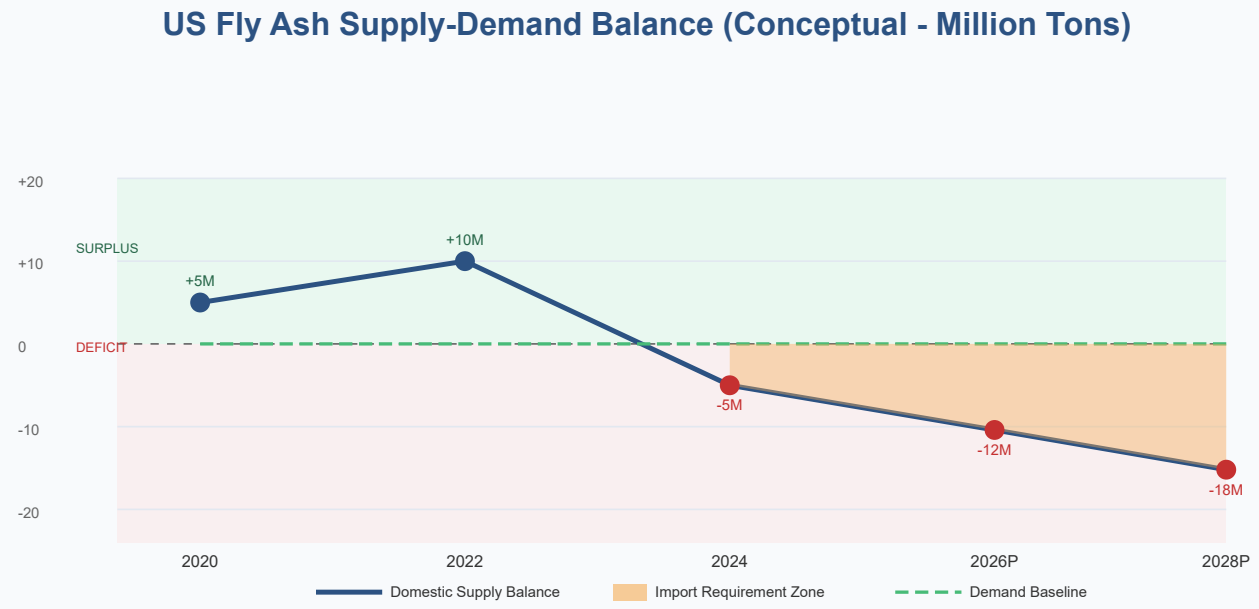
Source: ACAA Production & Use Surveys 2015-2023. Note: 2024 beneficial use rate reached 72% (record high).

2.1 Key Supply/Demand Observations

Year	CCP Production (M tons)	Beneficial Use (M tons)	Use Rate	Disposed (M tons)
2015	107	56	52%	51
2017	111	71	64%	40
2019	79	41	52%	38
2020	69	41	59%	28
2021	77	35	46%	42
2022	80	50	62%	30
2023	67	46	69%	21
2024 (est)	~60	~43	72%	~17

BENEFICIAL USE MILESTONE: 2024 marked the 10th consecutive year with >50% beneficial use rate. Record 72% rate achieved despite declining production. Over 4 million tons of previously disposed ash was reclaimed and utilized in 2023 - indicating markets are adapting to supply constraints.

3. SUPPLY-DEMAND GAP ANALYSIS



Conceptual projection based on coal plant retirements and steady concrete demand from IIJA spending. Gap must be filled by imports or reclaimed ash.

3.1 Supply Gap Drivers

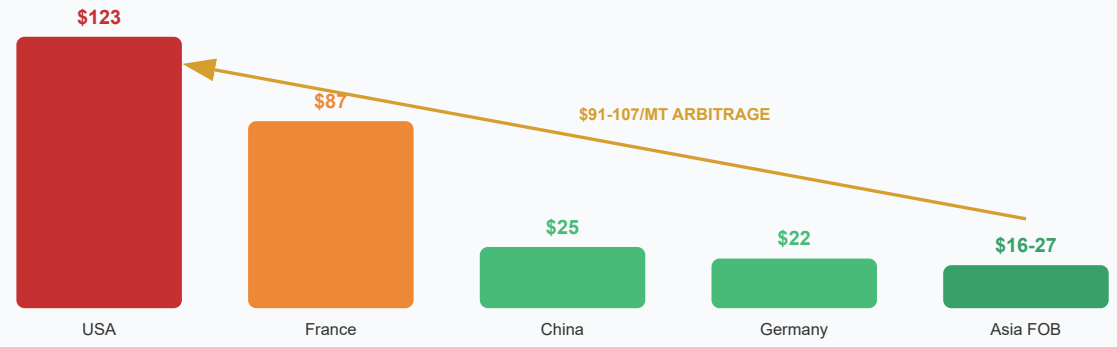
Supply Constraints

- **Coal Retirements:** 68.8 GW scheduled 2025-2030
- **Production Decline:** 107M tons (2015) → 67M tons (2023)
- **Plant Capacity Factor:** Dropped from 67% (2010) to 42% (2023)
- **Regional Concentration:** MISO + PJM losing 66% of 2025 retirements

Demand Drivers

- **IIJA Spending:** $1.2T through 2028 ($62B/year)
- **Cement Demand:** +4.5% annual growth through 2029
- **Low-Carbon Mandates:** California CALGreen, NY regulations
- **Record Utilization:** 72% beneficial use rate

4. IMPORT ECONOMICS & TRADE FLOW



Sources: ChemAnalyst, IMARC Group (Q2-Q3 2025). US prices +$5 YoY driven by tight supply and strong construction demand.

4.1 US Import Sources (Nov 2023 - Oct 2024)

Source Country	Share of Imports	Shipments	Notes
South Korea	49%	19	Largest supplier, high-quality Class F
Turkey	26%	10	Growing supplier
China	23%	9	145% tariff limits growth
Other (Japan, Canada)	2%	1	Alternative sources developing
Total	**100%**	**39**	**US = 11% of global fly ash imports**

4.2 Key Import States

Highest Volume Recipients: Florida, Washington, Louisiana, Nevada, Ohio, Texas

> **IMPORT ECONOMICS:** Asian FOB $16-27/MT + Freight (~$30-50/MT) = Landed cost $46-77/MT vs. US domestic $118-123/MT. Even with freight, imports offer 37-61% cost savings. Declining Asia-USWC freight rates through 2024-2025 further improve arbitrage.

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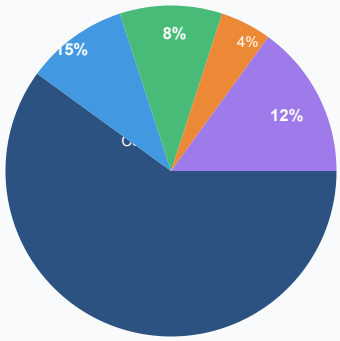
5. BENEFICIAL USE BY APPLICATION

2023 Fly Ash Beneficial Use by Application (46.3M tons total)



- Concrete/Grout (28.2M tons)
- Structural Fills (6.9M tons)
- Road Base/Stabilization (3.7M tons)
- Mine Backfill (1.9M tons)
- Other (Gypsum, Cement, etc.) (5.6M tons)

CONCRETE USE GROWING:
11.9M tons (2023) vs 10.9M tons (2022) = +9% YoY

Source: ACAA 2023 Production & Use Survey. Fly ash in concrete increased +9% YoY, reflecting infrastructure demand.

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6. MARKET PRICING & TRENDS

6.1 Current Pricing (2025)

$123/MT
USA (Q3 2025)
+$5 from 2024

$118/MT
USA (Q2 2025)
-5.3% QoQ

$87/MT
France
Stable

$22/MT
Germany
Oversupply

$16-27/MT
Asian FOB
Abundant supply

6.2 Price Drivers (2024-2025)

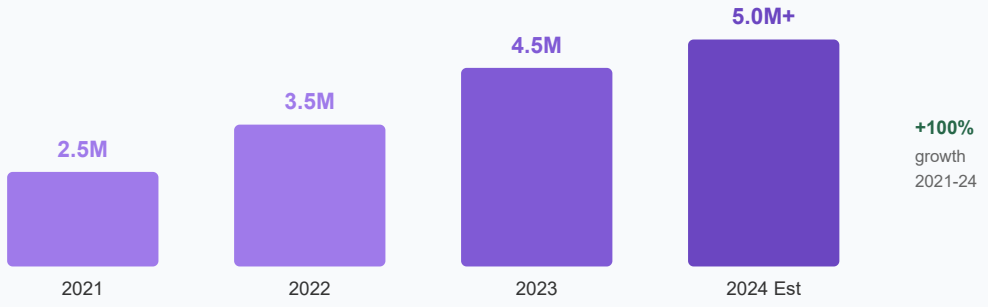
Factor	Impact	Direction
US Coal Plant Closures	Reduced domestic supply	Price Up

IIJA Infrastructure Spending	Sustained concrete demand	Price Up
145% China Tariff	Reduced Chinese imports	Price Up
Declining Asia-USWC Freight	Lower import costs	Price Down
Alternative Sources (Korea, Turkey)	Increased competition	Price Down
ILA Strikes / Port Congestion	Supply disruption (Nov 2024)	Price Up

7. RECLAIMED FLY ASH - EMERGING SUPPLY SOURCE

7.1 Reclamation Volume Trend

Previously Disposed Ash Reclaimed for Beneficial Use (Million Tons)



Source: ACAA. Reclaimed ash used for concrete products, cement kiln feed, pond closure activities. Markets adapting to supply decline.

7.2 Duke Beckjord Site Profile

Item	Details
Location	20 miles east of Cincinnati, OH (Ohio River)
Original Owner	Duke Energy (closed 2014)
Current Owner	Commercial Liability Partners (CLP) - purchased 2018
Site Size	1,400 acres
Coal Ash Volume	10+ billion pounds in unlined ponds
Regulatory Status	EPA CCR Rule compliance required; 100+ residents lawsuit pending

Potential	Port terminal development; ash reclamation opportunity

Reclaimed Ash: PROS

- Domestic supply as coal plants close
- Reduces environmental liability
- Strong market demand at premium prices
- Supports circular economy goals
- Long-term contracts available
- Potential REE recovery in future

Reclaimed Ash: CONS

- Variable quality - contamination risk
- High moisture requires drying
- May not meet ASTM C618 specs
- Beneficiation costs (LOI reduction)
- Regulatory uncertainty (EPA CCR)
- Community opposition/litigation

8. MARKET OUTLOOK SUMMARY

US Fly Ash Market - 5-Year Outlook (2025-2030)

Factor	Trend	Impact on Supply	Impact on Price
Coal Capacity	165 GW (2025) → 106 GW (2030): -36%	NEGATIVE	UP
CCP Production	67M tons (2023) → ~45M tons (2030 est)	NEGATIVE	UP
Beneficial Use Rate	72% (2024) → 80%+ (2030 target)	POSITIVE	NEUTRAL
Reclaimed Ash	4.5M tons (2023) → 8-10M tons (2030 est)	POSITIVE	DOWN
Import Volume	Growing - Korea, Turkey, Canada, Japan	POSITIVE	DOWN
IIJA Demand	$62B/year through 2028; concrete +4.5%/year	N/A	UP
Carbon Regulations	Buy Clean mandates; CALGreen Code expansion	N/A	UP

NET ASSESSMENT: US fly ash market will remain supply-constrained through 2030. Domestic production decline (-33% projected) will outpace demand moderation. Gap to be filled by: (1) increased imports from Asia/Turkey, (2) reclaimed ash from landfills/ponds, (3) alternative SCMs (slag, natural pozzolans). Prices likely to remain elevated ($100-130/MT range) with periodic spikes during supply disruptions.

9. QUICK REFERENCE - KEY NUMBERS

Metric	Value	Trend
US Price (Q3 2025)	$123/MT	Up $5 YoY
Asian FOB Price	$16-27/MT	Stable
Import Arbitrage	$91-107/MT	Attractive
US Coal Capacity (End 2025)	165 GW	-48% from 2011 peak
Coal Retirements 2025	14.5 GW	Record year
CCP Production (2023)	67M tons	-16% from 2022
Beneficial Use Rate (2024)	72%	Record high
Fly Ash in Concrete (2023)	11.9M tons	+9% YoY
Reclaimed Ash (2023)	4.5M tons	+29% YoY
Top Import Source	South Korea (49%)	Growing
US Market Size (2024)	$2.36 billion	+5.2% CAGR
IIJA Infrastructure	$1.2T through 2028	Demand driver

SOURCES

Industry Organizations: ACAA (American Coal Ash Association) Production & Use Surveys 2015-2024, ARTBA Forecast Update 2024
Government: EIA (Energy Information Administration) Coal Capacity Data, USGS CCP Statistics, EPA CCR Rule
Market Research: ChemAnalyst (pricing), IMARC Group, Fortune Business Insights, Market Data Forecast, Verified Market Reports
Energy Analysis: IEEFA (Institute for Energy Economics and Financial Analysis) - Coal Capacity Reports 2024-2025
Trade Data: Volza Import Statistics (Nov 2023 - Oct 2024)
Technical: ASTM C618, Power Engineering, Eco Material Technologies
News: Concrete Products Magazine, Utility Dive, Global Energy Monitor

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